第10章 文件操作 —— 课后作业答案

基础练习

1. 选择题

题号	答案	解析
(1)	**A** `fopen("data.txt", "r")`	`"r"` = 只读；`"w"` = 只写（会清空）；`"a"` = 追加；`"r+"` = 读写
(2)	**B**	`SEEK_END` 表示文件末尾，偏移量 0 表示定位到末尾

2. 判断题

题号	答案	解析
(1)	×	必须调用 `fclose` 关闭文件，否则可能导致数据丢失和资源泄漏
(2)	×	`"w"` 模式会清空原文件内容，若想保留原内容应使用 `"a"` 追加模式

编程练习

3. 文件复制

```c
#include <stdio.h>
#include <stdlib.h>

int main()
{
    FILE *src, *dest;
    char srcName[100], destName[100];
    char ch;

    printf("请输入源文件名: ");
    scanf("%s", srcName);
    printf("请输入目标文件名: ");
    scanf("%s", destName);

    src = fopen(srcName, "r");
    if (src == NULL)
    {
        printf("无法打开源文件: %s\n", srcName);
        return 1;
    }

    dest = fopen(destName, "w");
```

```
    if (dest == NULL)
    {
        printf("无法创建目标文件: %s\n", destName);
        fclose(src);
        return 1;
    }

    // 逐字符复制
    while ((ch = fgetc(src)) != EOF)
        fputc(ch, dest);

    printf("文件复制成功!\n");

    fclose(src);
    fclose(dest);
    return 0;
}
```

4. 统计文件字符数

```
#include <stdio.h>
#include <ctype.h>

int main()
{
    FILE *fp;
    char filename[100];
    char ch;
    int letters = 0, digits = 0, spaces = 0, others = 0;

    printf("请输入文件名: ");
    scanf("%s", filename);

    fp = fopen(filename, "r");
    if (fp == NULL)
    {
        printf("无法打开文件: %s\n", filename);
        return 1;
    }

    while ((ch = fgetc(fp)) != EOF)
    {
        if ((ch >= 'A' && ch <= 'Z') || (ch >= 'a' && ch <= 'z'))
            letters++;
        else if (ch >= '0' && ch <= '9')
            digits++;
        else if (ch == ' ' || ch == '\t' || ch == '\n')
            spaces++;
        else
            others++;
    }
```

```
    printf("字母: %d\n", letters);
    printf("数字: %d\n", digits);
    printf("空白(空格/制表/换行): %d\n", spaces);
    printf("其他: %d\n", others);
    printf("总计: %d\n", letters + digits + spaces + others);

    fclose(fp);
    return 0;
}
```

---

5. 文件加密

加密程序：

```
#include <stdio.h>

int main()
{
    FILE *in, *out;
    char inName[100], outName[100];
    char ch;

    printf("请输入要加密的文件名: ");
    scanf("%s", inName);
    printf("请输入加密输出文件名: ");
    scanf("%s", outName);

    in = fopen(inName, "r");
    out = fopen(outName, "w");

    if (in == NULL || out == NULL)
    {
        printf("文件打开失败!\n");
        return 1;
    }

    while ((ch = fgetc(in)) != EOF)
        fputc(ch + 1, out);   // ASCII码 +1 加密

    printf("加密完成! 加密文件: %s\n", outName);

    fclose(in);
    fclose(out);
    return 0;
}
```

解密程序：

```
#include <stdio.h>
```

```c
int main()
{
    FILE *in, *out;
    char inName[100], outName[100];
    char ch;

    printf("请输入加密文件名: ");
    scanf("%s", inName);
    printf("请输入解密输出文件名: ");
    scanf("%s", outName);

    in = fopen(inName, "r");
    out = fopen(outName, "w");

    if (in == NULL || out == NULL)
    {
        printf("文件打开失败!\n");
        return 1;
    }

    while ((ch = fgetc(in)) != EOF)
        fputc(ch - 1, out);   // ASCII码 -1 还原

    printf("解密完成! 解密文件: %s\n", outName);

    fclose(in);
    fclose(out);
    return 0;
}
```

---

6. 学生成绩文件读写

```c
#include <stdio.h>
#define N 10

typedef struct
{
    int id;
    char name[20];
    float s1, s2, s3;
    float total, avg;
} Student;

int main()
{
    Student stu[N];
    FILE *fp;

    // 输入学生数据
    printf("请输入%d个学生信息:\n", N);
    for (int i = 0; i < N; i++)
```

```
    {
        printf("第%d个 - 学号 姓名 三门成绩: ", i + 1);
        scanf("%d %s %f %f %f",
              &stu[i].id, stu[i].name,
              &stu[i].s1, &stu[i].s2, &stu[i].s3);
        stu[i].total = stu[i].s1 + stu[i].s2 + stu[i].s3;
        stu[i].avg = stu[i].total / 3.0;
    }

    // 写入原始数据文件
    fp = fopen("scores.dat", "wb");
    fwrite(stu, sizeof(Student), N, fp);
    fclose(fp);
    printf("已保存 %d 条记录到 scores.dat\n", N);

    // 从文件读取
    Student readStu[N];
    fp = fopen("scores.dat", "rb");
    fread(readStu, sizeof(Student), N, fp);
    fclose(fp);

    // 按总分排序（降序，简单冒泡）
    for (int i = 0; i < N - 1; i++)
        for (int j = 0; j < N - 1 - i; j++)
            if (readStu[j].total < readStu[j + 1].total)
            {
                Student t = readStu[j];
                readStu[j] = readStu[j + 1];
                readStu[j + 1] = t;
            }

    // 写入排名结果
    fp = fopen("rank.txt", "w");
    fprintf(fp, "====== 成绩排名 ======\n");
    fprintf(fp, "%-6s %-10s %-6s %-6s %-6s %-7s %-6s\n",
            "名次", "姓名", "课1", "课2", "课3", "总分", "平均");
    for (int i = 0; i < N; i++)
    {
        fprintf(fp, "%-6d %-10s %-6.1f %-6.1f %-6.1f %-7.1f %-6.1f\n",
                i + 1, readStu[i].name,
                readStu[i].s1, readStu[i].s2, readStu[i].s3,
                readStu[i].total, readStu[i].avg);
    }
    fclose(fp);
    printf("排名结果已保存到 rank.txt\n");
    return 0;
}
```

7. 日志记录

```
#include <stdio.h>
```

```c
#include <time.h>

void logMsg(const char *msg)
{
    FILE *fp;
    time_t now;
    struct tm *tm_info;
    char timestamp[30];

    // 获取当前时间
    time(&now);
    tm_info = localtime(&now);
    strftime(timestamp, sizeof(timestamp),
             "%Y-%m-%d %H:%M:%S", tm_info);

    // 追加写入日志文件
    fp = fopen("log.txt", "a");
    if (fp == NULL)
        return;

    fprintf(fp, "[%s] %s\n", timestamp, msg);
    fclose(fp);
}

int main()
{
    logMsg("系统启动");
    logMsg("用户 admin 登录成功");
    logMsg("执行数据备份操作");
    logMsg("用户 admin 退出");

    printf("日志已写入 log.txt\n");

    // 显示日志内容
    FILE *fp = fopen("log.txt", "r");
    if (fp)
    {
        printf("\n===== 日志内容 =====\n");
        char line[256];
        while (fgets(line, sizeof(line), fp))
            printf("%s", line);
        fclose(fp);
    }
    return 0;
}
```

运行后 `log.txt` 内容示例：

```
[2025-01-15 10:30:00] 系统启动
[2025-01-15 10:30:05] 用户 admin 登录成功
[2025-01-15 10:35:00] 执行数据备份操作
[2025-01-15 10:40:00] 用户 admin 退出
```

> **说明**：日志函数在任何需要记录的程序中都可调用，`"a"` 模式确保每次写入追加到文件末尾，不会覆盖历史日志。